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**Editor-in-Chief**

*Journal of Chemical Information and Modeling*  
American Chemical Society

**Subject: Submission of Original Research Article**

Dear Editor-in-Chief,

We are pleased to submit our manuscript entitled “**Multi-Objective MCTS for Antimalarial Design**” for consideration as a full Article in the *Journal of Chemical Information and Modeling*.

In this study, we present a novel generative architecture that integrates a Variational Autoencoder operating on molecular scaffolds (ScafVAE) with Monte Carlo Tree Search (MCTS) for the de novo design of antimalarial compounds. Generative chemistry frequently struggles to balance the exploration of vast chemical spaces with the exploitation of synthesizable, high-affinity regions. We address this by developing a dynamic progressive widening mechanism within the MCTS framework, guided by the ScafVAE learned priors, which biases the search tree toward chemically plausible fragment attachments.

We evaluate the generated candidates on a Pareto multi-objective frontier spanning multi-parameter optimization (MPO), synthetic accessibility (SA), and synthesizability (SYBA), with resistance-resilience (RRS) and polypharmacology-network (PNS) oracles linking de novo generation to experimental validation against drug-resistant *P. falciparum* strains. Across 20 independent seeds, our four-method benchmark shows that Random search (mean reward 0.6645) narrowly edges out MCTS+ScafVAE (0.6594, paired  $t_{19} = 2.32$ ,  $p = 0.032$ ), yet MCTS is the only method that systematically yields a diverse set of four non-dominated Pareto-optimal candidates (hypervolume 1.2366) spanning the potency–synthetic-accessibility trade-off—trade-offs that scalar optimization systematically misses.

Our  $2^5$  factorial ablation (160 runs) identifies the ScafVAE policy (+0.148), the Pareto front (+0.108), and a large fragment vocabulary (+0.079) as the dominant drivers of reward, while a fragment-vocabulary ablation confirms that restricting the vocabulary to aromatics alone severely degrades performance (ANOVA  $F = 350.1$ ,  $p = 1.12 \times 10^{-26}$ ). We believe our methodology offers a robust, generalizable framework for multi-objective molecular design and aligns perfectly with JCIM’s scope of advancing computational chemistry methods.

This manuscript is entirely original, has not been published previously, and is not under consideration for publication elsewhere. All authors have approved the manuscript and agree with its submission to JCIM.

We thank you for your time and consideration and look forward to the reviewers’ comments.

Sincerely,

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